Exploring Risk Gradients

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This document contains an attempt to explore how the relative contribution of NCEs to the total predator effect varies across a risk gradient. This is similar to Fig. 4 in the grant proposal, which uses predator density as a proxy for risk.

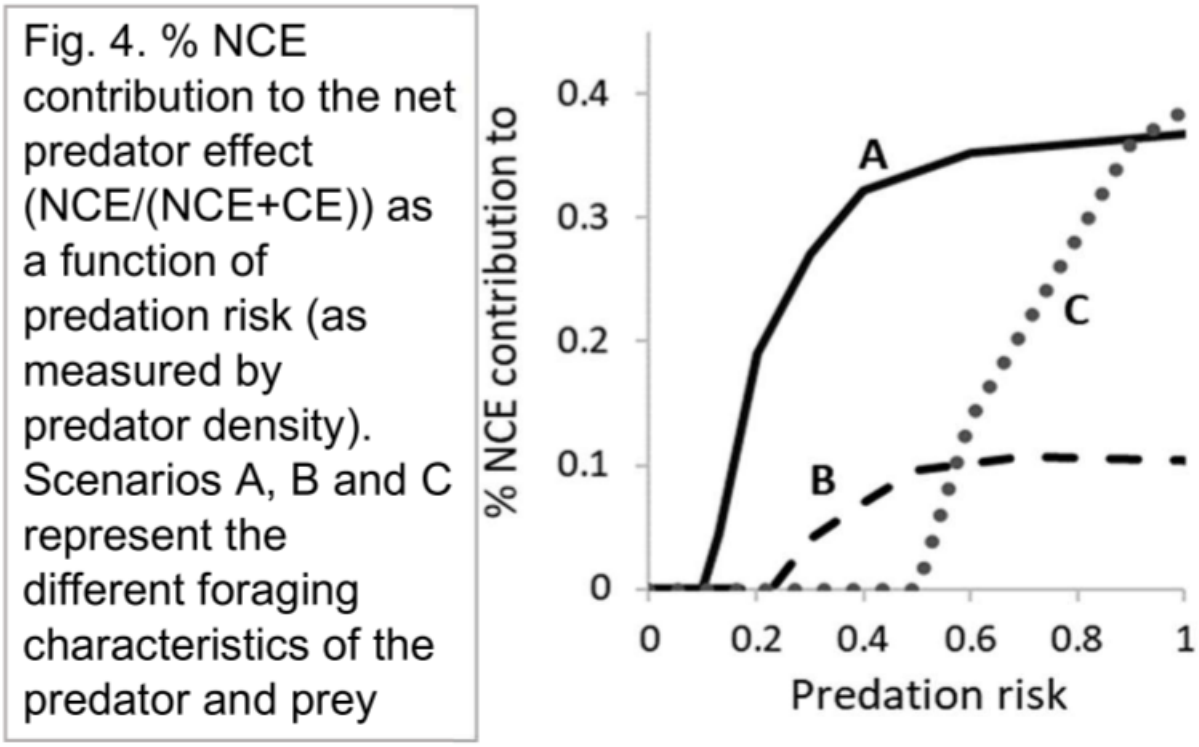


Figure 1: Fig. 4 from Grant Proposal

Notably, the curves generally reach a saturation point, where the contribution of NCEs to the total predator effect levels off with increasing risk. One goal of this document is determining why the percentage of the total predator effect composed of the NCE generally levels off given increasing risk.

Exploring the Mechanisms

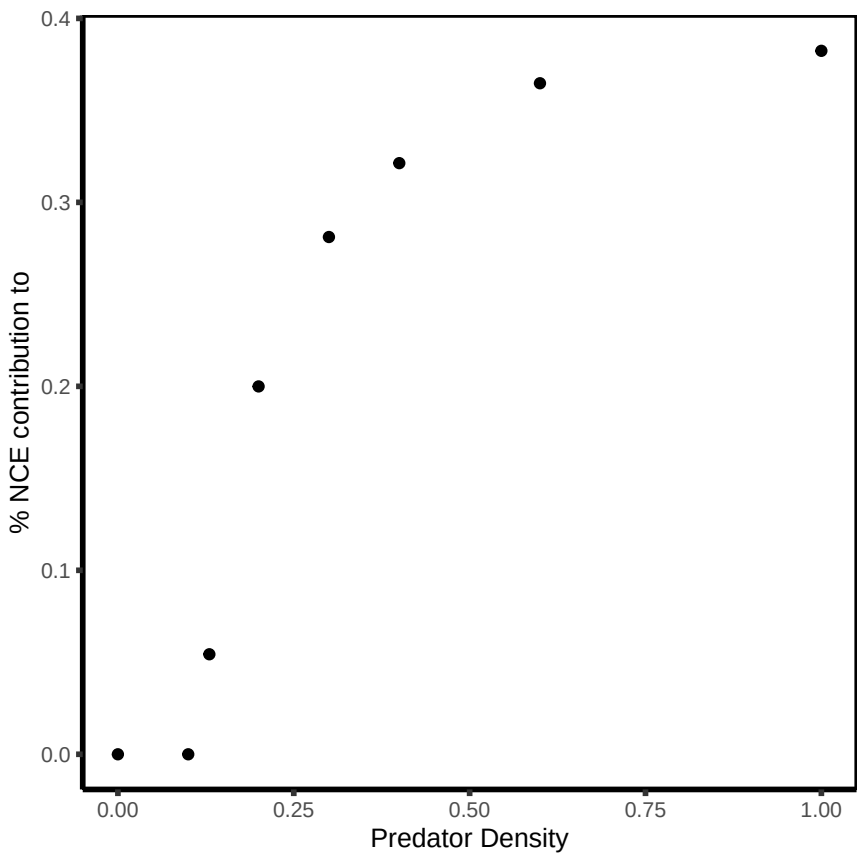
We will first recreate curve A in the above figure, but on a finer scale. Specifically, we will look at the underlying IREF curves and their shapes to determine why this relationship is saturating.

```
library(ggplot2)
library(dplyr)
```

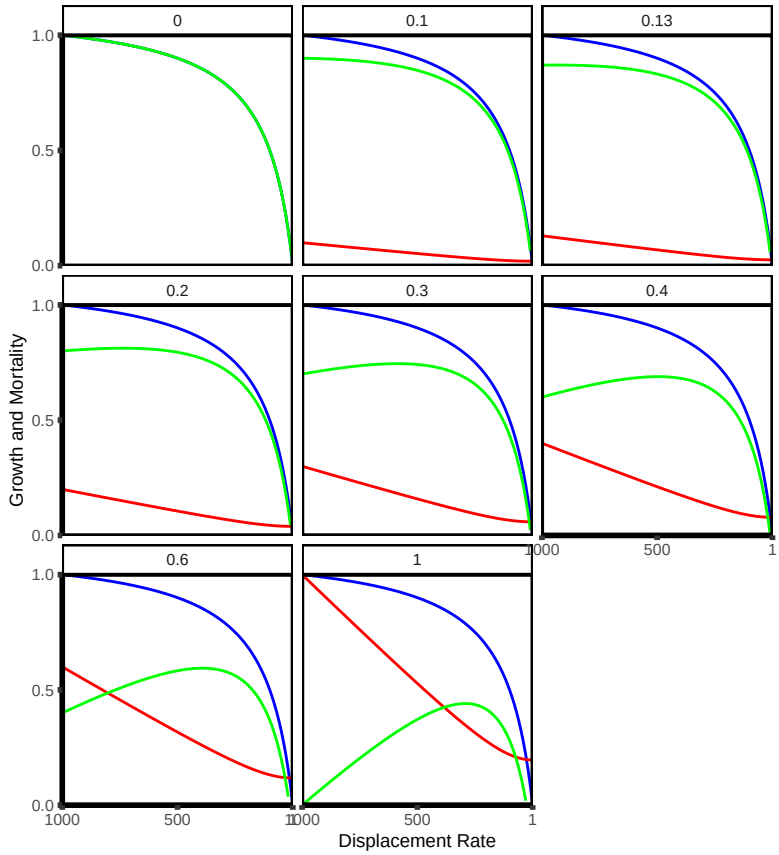
```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# Setting the range of prey displacement rate
trait.vec <- seq(0, 1, by = 0.01)
speed.vec <- 1 - trait.vec

# These formulas are simplified by setting some of the values as constants
# r is always assumed to be 1, while k, s, and v are manipulated values
pred.curve <- function(k, v, s){
  k*sqrt(v^2 + s^2)
}

# R is always assumed to be 1, while s and h are manipulated values
growth.curve <- function(s, R, h){
  (R * s)/(1 + s*R*h)
}

k_vec <- c(0, 0.1, 0.13, 0.2, 0.3, 0.4, 0.6, 1) # Range of values used by Scott for Fig. 4 in the grant
h_val <- 8
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
pred_constant_val <- 0.98058

# Generate growth and mortality curves for each value of predator density (k_vec)
fig4.df <- as.data.frame( mat.or.vec( length( trait.vec ) * length(k_vec), 7) )
colnames(fig4.df) <- c("trait", "speed", "growth_curve", "mort_curve", "fitness", "pct_NCE", "pred_dens

row <- 0

for (i in 1:length(k_vec)){
  for (j in 1:length(trait.vec)){
    row <- row + 1
    fig4.df[row, 1] <- trait.vec[j]
    fig4.df[row, 2] <- speed.vec[j]
```

```
    fig4.df[row, 3] <- cost_constant_val * growth.curve( s = fig4.df[row, 2], R = R_val, h = h_val )
    fig4.df[row, 4] <- pred_constant_val * pred.curve( k = k_vec[i], v = v_val, s = fig4.df[row, 2] )
    fig4.df[row, 5] <- fig4.df[row, 3] - fig4.df[row, 4]
    fig4.df[row, 6] <- (1 - fig4.df[row, 3])/((1 - fig4.df[row, 3]) + fig4.df[row, 4])
    fig4.df[row, 7] <- k_vec[i]
  }
}
```



First, NCE contributions to the total predator effect, to ensure that this was done similarly to Fig. 4 in the grant proposal above. This shows the same saturating relationship, which is good news. Next, we'll look under the hood at the underlying IREF curves.



The above figure shows the growth (in blue), mortality (in red), and fitness (in green) curves, as a function of prey displacement rate. Each plot represents a different predator density, noted in the header above the plot.

The reason that the NCE's contribution to the total predator effect tends to level off with increasing risk seems to occur for two main reasons.

The optimal RITR seems to occur roughly around the inflection point of the growth curve, except when predator density (and therefore risk) is low ($k > 0.2$) and a weak response is expected by the prey.

Put another way, as risk increases, there is eventually a point at which the benefit of the RITR (in terms of reduction of mortality due to predation) becomes too costly. Quantitatively, this is probably where the slope of the growth curve exceeds greatly exceeds the slope of the mortality curve.

This finding is probably general in cases where the growth curve is concave-down, while predation risk declines in a linear or saturating way.

Here is a table showing the optimal trait and speed values, and the percentage of the total predator effect consisting of NCEs, as a function of predator density.

```
optim.df[ -c(3:5) ]
```

```
## # A tibble: 8 x 4
## # Groups:   pred_dens [8]
##   trait speed pct_NCE pred_dens
##   <dbl> <dbl>   <dbl>     <dbl>
## 1  0     1     0           0
## 2  0     1     0           0.1
## 3  0.06  0.94  0.0544      0.13
## 4  0.26  0.74  0.200       0.2
## 5  0.41  0.59  0.281       0.3
## 6  0.5   0.5   0.321       0.4
## 7  0.61  0.39  0.365       0.6
## 8  0.71  0.29  0.382       1
```

Incidentally, the inflection point of the growth curve is at around a displacement rate of 375 (calculated below using the R package 'inflection').

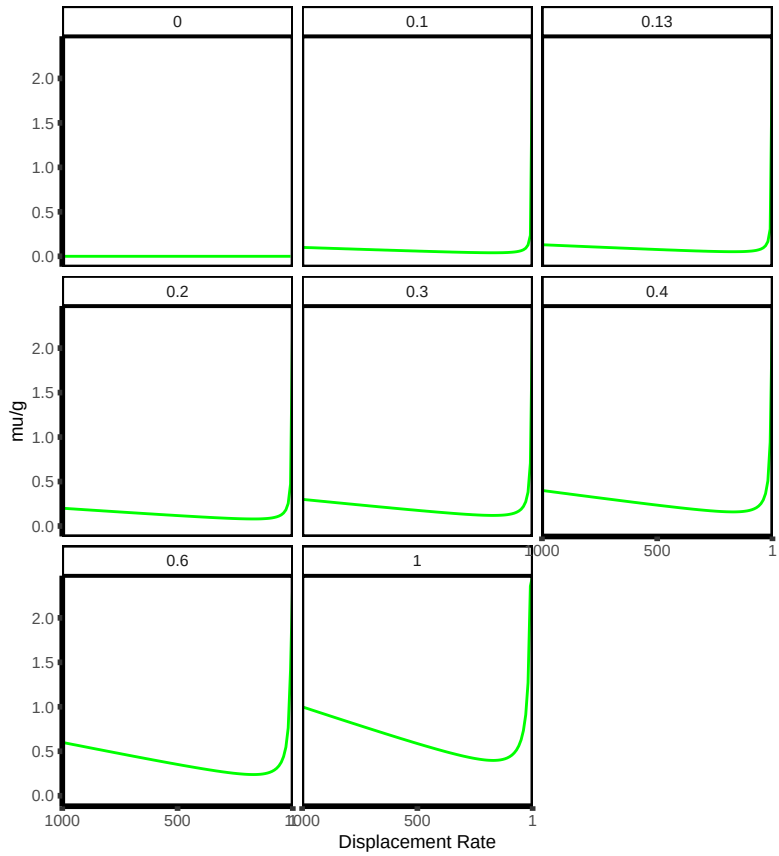
```
## $ctype
## [1] "convex"
##
## $index
## [1] 0
```

```
##     j1 j2   chi
## EDE  1 76 0.625
```

The inflection point lies at the value labeled "chi". Note that this is the trait value, so to get the displacement rate you must subtract it from 1, then multiply by 1000.

Other Fitness Optimization Functions

Out of curiosity, what if we had chosen to optimize fitness in the above example using μ/g? Do we see the same general behavior of the optimum, and the proportion of the total predator effect that comes from NCEs?



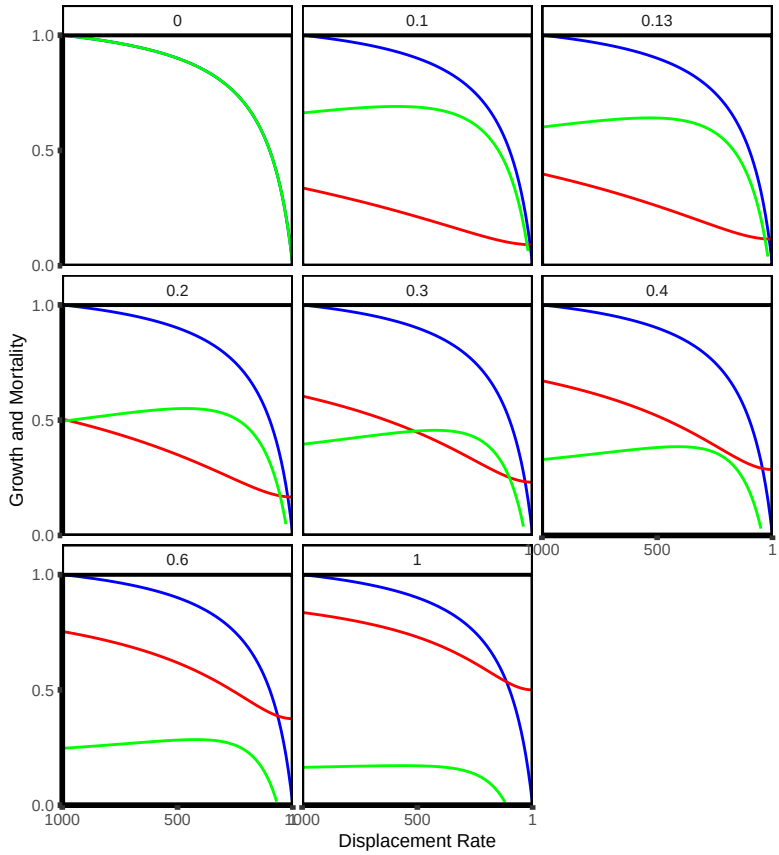

```
## # A tibble: 8 x 4
## # Groups:   pred_dens [8]
##   trait speed pred_dens  mu_g
##   <dbl> <dbl> <fct>     <dbl>
## 1  0     1    0         0
## 2  0     1    0.1       0.100
## 3  0.06  0.94 0.13      0.123
## 4  0.26  0.74 0.2       0.156
## 5  0.41  0.59 0.3       0.197
## 6  0.5   0.5  0.4       0.235
## 7  0.61  0.39 0.6       0.303
## 8  0.71  0.29 1         0.439
```

The optimal displacement rate (2nd column) is exactly the same as before.

What if the Predator has a Type II FR?

Last, I want to explore whether the above finding would change if the predator had a Type II functional response, as is typical for most predators. This may help to establish whether this finding is "general" to the majority of predator-prey systems. We'll start with an (arbitrary) predator handling time of 5.

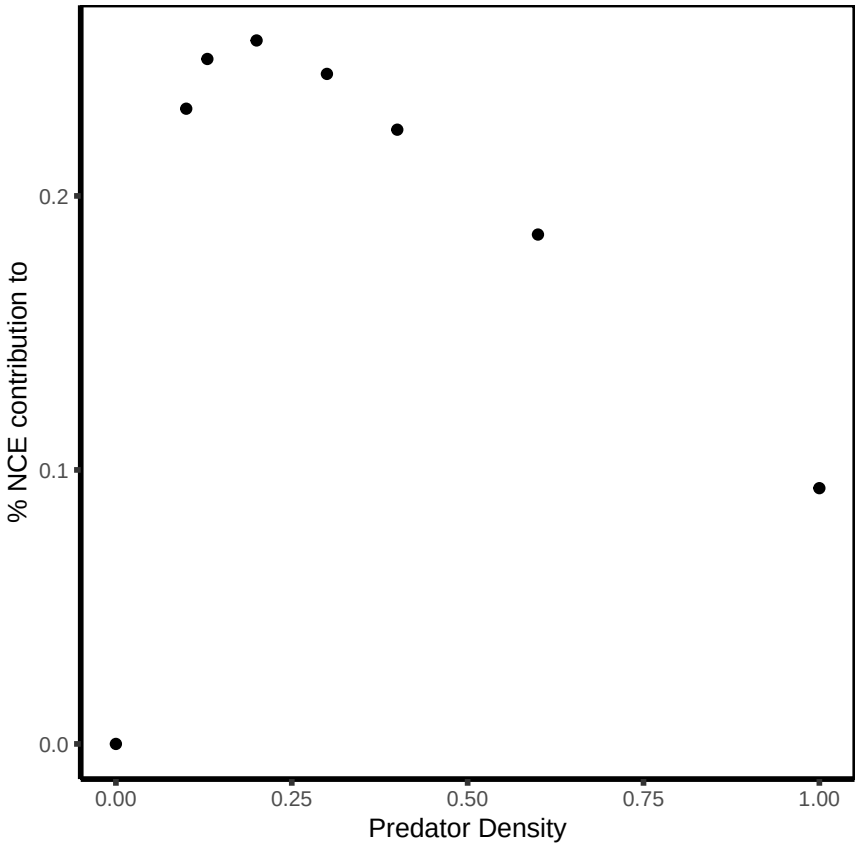
```
# First, write a new Type II mortality curve
# This function will take the above encounter rate formula and apply handling time to it
typeII.curve <- function( a_pred, h_pred ){
  a_pred / ( 1 + a_pred * h_pred )
}

k_vec <- c(0, 0.1, 0.13, 0.2, 0.3, 0.4, 0.6, 1) # Range of values used by Scott for Fig. 4 in the grant
h_val <- 8
h_pred_val <- 5 # Start at a predator handling time value of 5
R_val <- 1
cost_constant_val <- 9
v_val <- 0.2
pred_constant_val <- 5

# Generate growth and mortality curves for each value of predator density (k_vec)
typeII.df <- as.data.frame( mat.or.vec( length( trait.vec ) * length(k_vec), 7) )
colnames(typeII.df) <- c("trait", "speed", "growth_curve", "mort_curve", "fitness", "pct_NCE", "pred_den

row <- 0

for (i in 1:length(k_vec)){
  for (j in 1:length(trait.vec)){
    row <- row + 1
    typeII.df[row, 1] <- trait.vec[j]
    typeII.df[row, 2] <- speed.vec[j]
    typeII.df[row, 3] <- cost_constant_val * growth.curve( s = typeII.df[row, 2], R = R_val, h = h_val
    typeII.df[row, 4] <- pred_constant_val * typeII.curve( a_pred = pred.curve( k = k_vec[i], v = v_val
    typeII.df[row, 5] <- typeII.df[row, 3] - typeII.df[row, 4]
    typeII.df[row, 6] <- (1 - typeII.df[row, 3])/((1 - typeII.df[row, 3]) + typeII.df[row, 4])
    typeII.df[row, 7] <- k_vec[i]
  }
}
```



The behavior of the fitness curves is similar looking…

```
## # A tibble: 8 x 4
## # Groups:   pred_dens [8]
##   trait speed pct_NCE pred_dens
##   <dbl> <dbl>   <dbl>     <dbl>
## 1  0     1     0           0
## 2  0.41  0.59  0.232       0.1
## 3  0.47  0.53  0.250       0.13
## 4  0.54  0.46  0.257       0.2
## 5  0.58  0.42  0.245       0.3
## 6  0.59  0.41  0.224       0.4
## 7  0.58  0.42  0.186       0.6
## 8  0.43  0.57  0.0933      1
```

And the optimums are for the most part in a similar place (except at very high predator densities!), but…



The relative contribution of the NCE to the total predator effect has a very different behavior. It peaks when predator density is low, and then declines with increasing predator density. I think this is quantitatively because the position of the mortality curve as a whole – but especially at the far right end of the plots (i.e., where displacement rate is very low) – shifts upwards as predator density increases, which will always increase the impact of CEs.

Biologically, I think this is because at very low prey displacement rates, predation rates start to increase again because of the effect of handling time. This acts to push the optimal displacement rate back to the left, because if encounters with predators are infrequent enough, those predators will always be hungry.

Last, we'll revisit the above finding from the Type I FRs. To summarize, increasing predation risk drives the optimum to the right, but this rightward shift is limited by the costs to growth becoming too steep. That seems to be the case here too, but with an added wrinkle: if risk is sufficiently high and encounters with the predator become too infrequent, that can shift the optimal displacement rate back to the left.